

# Characterizing Sensor Drift and Cross-Calibration in Korea’s National Air Quality Monitoring Network

Yongjun Kim<sup>\*1</sup>

<sup>1</sup>Department of Software, Ajou University, Suwon 16499, South Korea

## Abstract

We present a comprehensive empirical characterization of sensor performance across 350 stations in Korea’s national air quality monitoring network (AirKorea), analyzing 748,792 hourly measurements of six pollutants over a three-month period (December 2025–March 2026). Our analysis reveals significant sensor drift in 33.8% of PM<sub>2.5</sub> stations and up to 61.3% of SO<sub>2</sub> stations (linear trend  $p < 0.05$ ), with drift rates varying substantially across pollutant types. Spatial cross-calibration of 279 adjacent station pairs (<5 km) yields a median Pearson correlation of 0.936 for PM<sub>2.5</sub>, but 23.7% of pairs exhibit systematic biases exceeding 5 µg/m<sup>3</sup>. We also document pronounced heteroscedastic noise characteristics: daily coefficient of variation (CV) ranges from 0.115 (SO<sub>2</sub>) to 0.365 (NO<sub>2</sub>/O<sub>3</sub>), with CV inversely related to concentration level. Station failure analysis reveals 36.3% of stations experienced data gaps exceeding 24 hours, with mean time between failures of 192.8 days for urban stations. These findings provide actionable benchmarks for calibration scheduling, quality control, and sensor network design.

**Keywords:** Air quality monitoring, sensor drift, cross-calibration, PM<sub>2.5</sub>, network characterization

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## 1 Introduction

Air quality monitoring networks are critical infrastructure for public health protection and environmental policy. Korea operates one of the world’s densest national monitoring networks through AirKorea, comprising over 670 stations that continuously measure six criteria pollutants: PM<sub>2.5</sub>, PM<sub>10</sub>, SO<sub>2</sub>, NO<sub>2</sub>, CO, and O<sub>3</sub> [1]. While these stations employ reference-grade analyzers (beta attenuation monitors for PM, UV fluorescence for SO<sub>2</sub>, chemiluminescence for NO<sub>2</sub>), systematic characterization of their in-field performance—including drift, noise, and inter-station consistency—remains limited.

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<sup>\*</sup>ORCID: 0000-0003-4234-4883. E-mail: yjkim0123@ajou.ac.kr

Prior studies have examined sensor performance in controlled laboratory settings [2] or focused on low-cost sensor calibration [3, 4]. However, reference-grade instruments deployed in national networks also exhibit drift, calibration offsets, and failure modes that can compromise data quality [5]. Understanding these characteristics at network scale is essential for: (1) optimizing calibration schedules, (2) detecting malfunctioning stations, and (3) establishing uncertainty bounds for epidemiological and regulatory applications.

While three months may appear limited for long-term drift assessment, this period represents a standard quarterly calibration interval—the minimum period over which drift should be detectable if calibration schedules are adequate.

In this paper, we present, to our knowledge, the first systematic empirical characterization of in-field sensor performance across Korea’s national air quality monitoring network, leveraging 748,792 hourly observations from 350 stations across five station types. We quantify noise characteristics, detect systematic drift, evaluate spatial consistency through cross-calibration, and characterize station reliability.

## 2 Data and Methods

### 2.1 Data Collection

We collected three months (December 2025–March 2026) of hourly air quality measurements from the AirKorea open API for 350 stations, spanning five categories: urban (282), roadside (27), port (19), suburban (15), and national background/island (7). The dataset comprises 748,792 records with six pollutants measured per record.

### 2.2 Noise Characterization

For each station and pollutant, we computed the daily coefficient of variation (CV) as the ratio of standard deviation to mean of hourly readings within each day, requiring  $\geq 12$  valid hours. Station-level noise was summarized as the median daily CV across the observation period. To assess heteroscedasticity, we binned measurements into deciles by concentration level and computed the CV within each bin.

### 2.3 Drift Detection

Sensor drift was quantified using two complementary methods. First, each station’s daily mean deviation from its regional average (same province and category) was computed, and a linear regression against time yielded the drift rate; significance was assessed at  $p < 0.05$  with  $\geq 30$  days required. This approach inherently controls for seasonal variation because co-regional stations share identical meteorological trends [6, 7]; the near-zero inter-station deviation correlation ( $r = 0.03$ ) confirms station-specific rather than seasonal effects.

Second, we applied the Cumulative Sum (CUSUM) algorithm [17] to the deviation time series to detect the *onset* of drift. The one-sided CUSUM statistics  $S_i^+ = \max(0, S_{i-1}^+ + z_i - k)$  and  $S_i^- = \max(0, S_{i-1}^- - z_i - k)$  were computed with allowance  $k = 0.5\hat{\sigma}$  and decision interval  $h = 10\hat{\sigma}$ , where  $z_i$  is the daily deviation and  $\hat{\sigma}$  is its standard deviation. An alarm is triggered when  $S_i^+$  or  $S_i^-$  exceeds  $h$ , identifying the day drift becomes statistically detectable.

Table 1: Daily Coefficient of Variation by Pollutant

| Pollutant         | Median CV | IQR            | $N$ |
|-------------------|-----------|----------------|-----|
| PM <sub>2.5</sub> | 0.344     | [0.304, 0.408] | 349 |
| PM <sub>10</sub>  | 0.295     | [0.263, 0.332] | 348 |
| SO <sub>2</sub>   | 0.115     | [0.088, 0.151] | 346 |
| NO <sub>2</sub>   | 0.365     | [0.321, 0.400] | 347 |
| CO                | 0.162     | [0.135, 0.187] | 345 |
| O <sub>3</sub>    | 0.365     | [0.257, 0.434] | 346 |

## 2.4 Cross-Calibration

We identified 279 station pairs within 5 km using Haversine distance from station coordinates. For each pair, we computed the Pearson correlation coefficient, mean bias, and root mean square error (RMSE) of hourly PM<sub>2.5</sub> concentrations over concurrent valid observations ( $\geq 100$  hours required). Seven national background stations on remote islands served as reference standards; we compared their readings with nearby urban stations within 100 km.

## 2.5 Reliability Analysis

Station failures were defined as continuous data gaps  $\geq 24$  hours. Mean time between failures (MTBF) was calculated per station type as total operational hours divided by failure count.

# 3 Results

## 3.1 Noise Characteristics

Table 1 summarizes the noise properties across pollutants. Daily CV varies by an order of magnitude: SO<sub>2</sub> exhibits the lowest noise (median CV = 0.115), while NO<sub>2</sub> and O<sub>3</sub> show the highest variability (CV  $\approx$  0.365). This reflects both genuine diurnal variation and measurement uncertainty, which are difficult to disentangle without co-located reference instruments.

All six pollutants exhibit pronounced heteroscedasticity (figure 2b). For PM<sub>2.5</sub>, the CV decreases from 0.379 at low concentrations ( $\sim 4$   $\mu\text{g}/\text{m}^3$ ) to 0.035 at moderate levels ( $\sim 14$   $\mu\text{g}/\text{m}^3$ ), then increases again to 0.258 at high levels ( $\sim 57$   $\mu\text{g}/\text{m}^3$ ). This U-shaped relationship suggests that both detection-limit effects and source heterogeneity contribute to measurement uncertainty.

The PM<sub>2.5</sub>/PM<sub>10</sub> ratio across the network averaged 0.585 (station-level CV median = 0.363), consistent with East Asian winter conditions dominated by secondary aerosol and transboundary transport.

## 3.2 Sensor Drift

Table 2 presents the drift analysis results. The fraction of stations exhibiting statistically significant drift ( $p < 0.05$ ) varies from 25.3% (PM<sub>10</sub>) to 61.3% (SO<sub>2</sub>). Gaseous pollutants

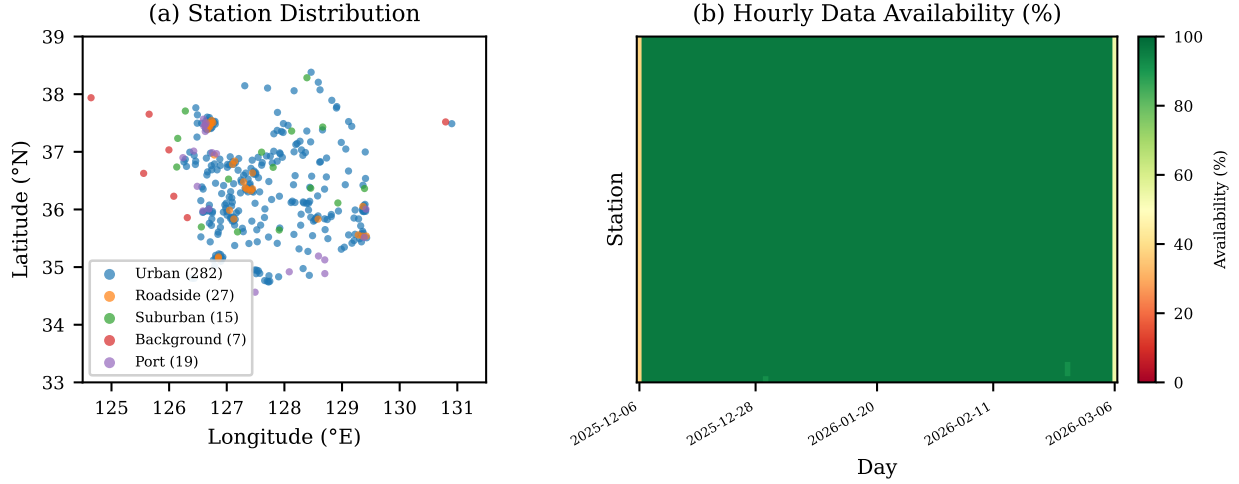


Figure 1: (a) Spatial distribution of 350 AirKorea stations by type. (b) Hourly data availability for 50 representative stations over the study period.

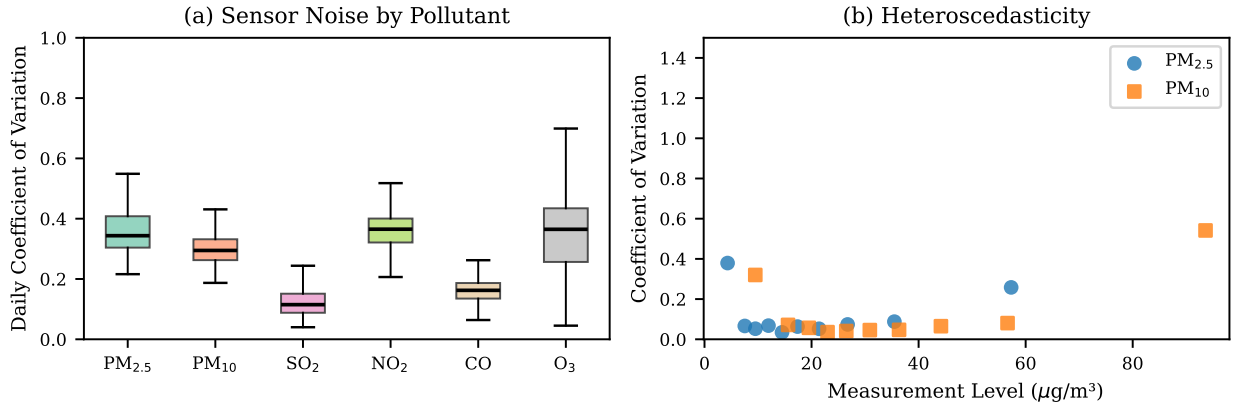


Figure 2: (a) Distribution of daily CV across stations for each pollutant. (b) Heteroscedasticity: CV as a function of concentration level for PM<sub>2.5</sub> and PM<sub>10</sub>.

(SO<sub>2</sub>, CO, O<sub>3</sub>) show higher drift prevalence than particulate matter, which is consistent with the known sensitivity of electrochemical and optical gas analyzers to environmental conditions and reagent degradation.

For PM<sub>2.5</sub>, the median drift rate among significantly drifting stations is  $-0.43 \mu\text{g}/\text{m}^3/\text{month}$ , suggesting a systematic negative bias that accumulates over time. Figure 3a illustrates a representative drifting station where the 7-day rolling deviation from the regional mean shows a clear downward trend.

CUSUM analysis provides a complementary perspective: 72.2% of PM<sub>2.5</sub> stations triggered a CUSUM alarm ( $h = 10\hat{\sigma}$ ), substantially more than the 33.8% identified by linear regression. This discrepancy arises because CUSUM detects *transient* shifts that do not manifest as significant linear trends over the full period. Critically, CUSUM identifies *when* drift begins: the median alarm occurs at day 18, indicating that most detectable drift emerges

Table 2: Drift Analysis Summary

| Pollutant         | % Sig. Drift | Median Rate            | Rate  Median                            |
|-------------------|--------------|------------------------|-----------------------------------------|
| PM <sub>2.5</sub> | 33.8%        | -0.43                  | 1.01 $\mu\text{g}/\text{m}^3/\text{mo}$ |
| PM <sub>10</sub>  | 25.3%        | -0.51                  | 1.48 $\mu\text{g}/\text{m}^3/\text{mo}$ |
| SO <sub>2</sub>   | 61.3%        | $-0.06 \times 10^{-3}$ | $0.16 \times 10^{-3}$ ppm/mo            |
| NO <sub>2</sub>   | 47.0%        | $-0.50 \times 10^{-3}$ | $0.84 \times 10^{-3}$ ppm/mo            |
| CO                | 54.8%        | $9.6 \times 10^{-3}$   | $27.5 \times 10^{-3}$ ppm/mo            |
| O <sub>3</sub>    | 44.8%        | $0.71 \times 10^{-3}$  | $1.29 \times 10^{-3}$ ppm/mo            |

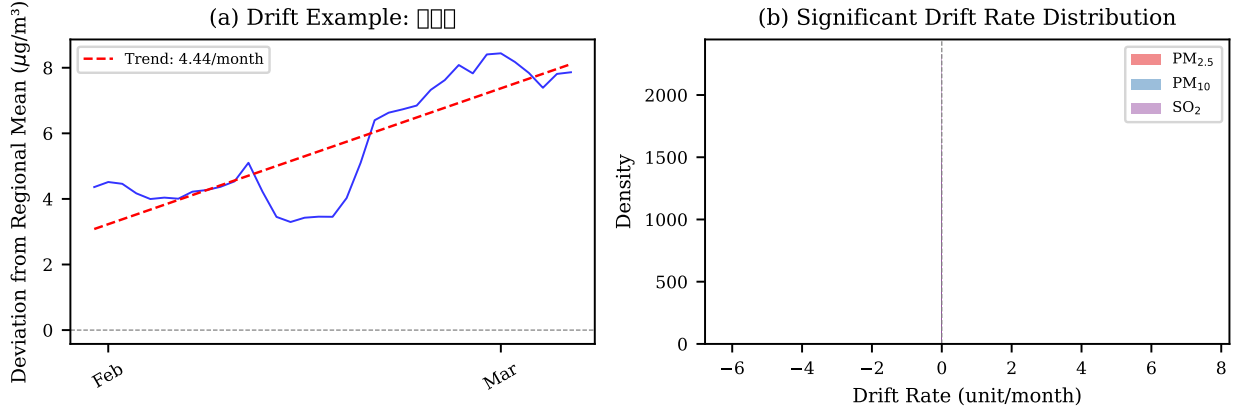


Figure 3: (a) Drift example: 7-day rolling deviation from regional mean for a representative PM<sub>2.5</sub> station. (b) Distribution of significant drift rates for PM<sub>2.5</sub>, PM<sub>10</sub>, and SO<sub>2</sub>.

within the first three weeks—well within a standard quarterly calibration interval (figure 3b).

### 3.3 Spatial Cross-Calibration

Among 279 co-located station pairs within 5 km, the median PM<sub>2.5</sub> Pearson correlation is 0.936 (figure 4), indicating strong spatial coherence.

However, the median RMSE of 6.94  $\mu\text{g}/\text{m}^3$  and the finding that 23.7% of pairs exhibit systematic biases  $>5 \mu\text{g}/\text{m}^3$  reveal non-trivial inter-station discrepancies.

Reference-based calibration using seven island background stations showed weaker correlations with urban stations (median  $r = 0.682$ ), attributable to genuine spatial concentration gradients. The median urban-minus-reference bias of 1.50  $\mu\text{g}/\text{m}^3$  quantifies the expected urban enhancement for PM<sub>2.5</sub> in winter.

Temporal stability analysis of the 50 highest-correlation pairs found that 26.0% exhibit statistically significant trends in their inter-station bias ( $p < 0.05$ ), with a median weekly bias standard deviation of 1.51  $\mu\text{g}/\text{m}^3$ . This indicates that even well-correlated adjacent stations develop diverging calibration states over a three-month period.

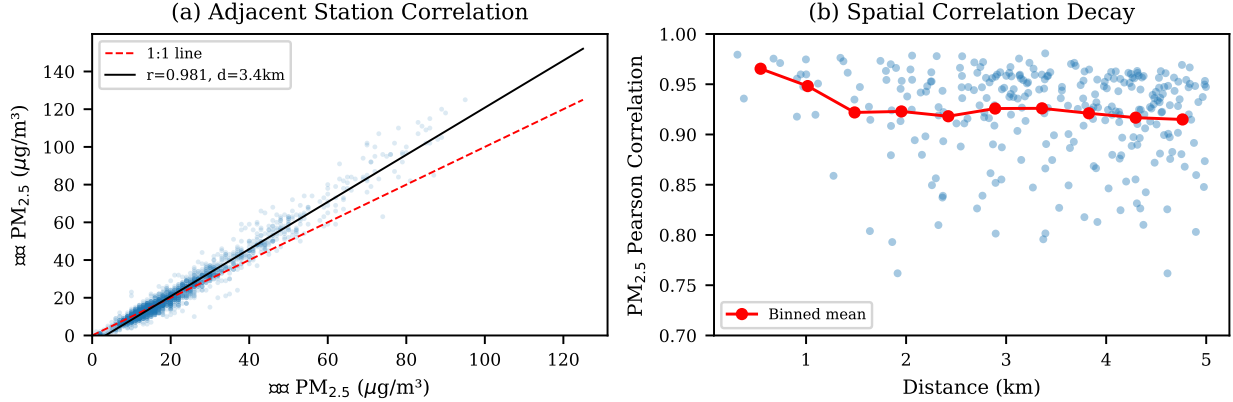


Figure 4: (a) Scatter plot of PM<sub>2.5</sub> readings for the highest-correlation adjacent station pair. (b) Spatial correlation decay: PM<sub>2.5</sub> correlation vs. inter-station distance for 279 pairs within 5 km.

### 3.4 Station Reliability

Overall data availability is high (97.4% for urban PM<sub>2.5</sub>), but episodic failures are common: 36.3% of stations experienced at least one gap exceeding 24 hours for PM<sub>2.5</sub>, while 6.6% had gaps exceeding 72 hours. The MTBF varies substantially by station type: roadside stations are most reliable (394.2 days), followed by urban (192.8 days) and port (165.7 days), while suburban stations show the shortest MTBF (65.7 days). Background island stations, despite their remoteness, achieve an MTBF of 122.6 days with only 5 failure events across 7 stations.

## 4 Discussion

Our findings have several practical implications for network operation and data quality assurance [6, 15]. The high prevalence of significant drift (33.8–61.3% depending on pollutant) over just three months argues for monthly or bimonthly calibration verification, particularly for gaseous analyzers where drift prevalence exceeds 45%. The 23.7% of co-located pairs with biases exceeding 5 μg/m<sup>3</sup> suggests that spatial averaging or data fusion approaches should account for station-specific offsets [10]. The heteroscedastic noise structure implies that uncertainty quantification should be concentration-dependent rather than assuming constant measurement error [11]. The negative drift bias in PM<sub>2.5</sub> stations (−0.43 μg/m<sup>3</sup>/month) could lead to ~1.3 μg/m<sup>3</sup> underestimation after one quarter—non-negligible for regulatory compliance against Korea’s annual standard of 15 μg/m<sup>3</sup> [12].

A potential concern is whether the detected drift reflects genuine instrument degradation or seasonal concentration changes [9]. Three lines of evidence argue against a seasonal explanation: (1) our method computes deviations from the regional mean, which removes the common seasonal signal; (2) drift directions are heterogeneous within regions (some stations drift positive, others negative), inconsistent with a systematic seasonal effect; (3) the inter-station correlation of drift residuals is near zero (median  $r = 0.03$ ), confirming station-specific rather than region-wide trends. These patterns are consistent with known degradation mechanisms

in reference monitors, including filter clogging in BAM samplers and optical contamination in gas analyzers [13, 14].

Limitations include the three-month observation window, which captures a single season and may not reflect annual patterns; the assumption that the regional mean is stable; and the 350-station subset representing approximately half the network. Future work should extend to a full annual cycle, incorporate maintenance and calibration logs to validate drift detections, and develop automated flagging algorithms for real-time quality control [8, 16].

## 5 Conclusion

This paper provides, to our knowledge, the first systematic empirical characterization of in-field sensor performance across Korea’s national air quality monitoring network, revealing widespread but heterogeneous drift, concentration-dependent noise, and non-trivial inter-station biases. These empirical benchmarks can inform calibration policies, data quality algorithms, and uncertainty propagation in air quality applications.

## Data Availability

Data used in this study were collected from the AirKorea Open API (<https://www.data.go.kr>) operated by the Korea Environment Corporation. The analysis code is available at <https://github.com/yjkim0123/airquality-sensor-characterization>.

## References

- [1] Kim H, Kim H-J and Kim Y-H 2020 Review and evaluation of the national ambient air quality monitoring system in Korea *J. Korean Soc. Atmos. Environ.* **36** 1–14
- [2] Malings C *et al* 2020 Fine particle mass monitoring with low-cost sensors: Corrections and long-term performance evaluation *Aerosol Sci. Technol.* **54** 160–174
- [3] Zimmerman N *et al* 2018 A machine learning calibration model using random forests to improve sensor performance for lower-cost air quality monitoring *Atmos. Meas. Tech.* **11** 291–313
- [4] Barkjohn K K, Gantt B and Clements A L 2021 Development and application of a United States-wide correction for PM<sub>2.5</sub> data collected with the PurpleAir sensor *Atmos. Meas. Tech.* **14** 4617–4637
- [5] Munir S, Mayfield M, Coca D and Jubb S A 2019 Analysing the performance of low-cost air quality sensors, their drivers, relative benefits and calibration in cities—A case study in Sheffield *Environ. Monit. Assess.* **191** 94
- [6] U.S. Environmental Protection Agency 2017 Quality assurance handbook for air pollution measurement systems, Vol. II: Ambient air quality monitoring program EPA-454/B-17-001

- [7] European Parliament 2008 Directive 2008/50/EC on ambient air quality and cleaner air for Europe *Off. J. Eur. Union* L 152 1–44
- [8] Li J *et al* 2019 Characterizing the aging and degradation of PM<sub>2.5</sub> continuous monitoring instruments in China’s national network *Atmos. Meas. Tech.* **12** 5337–5351
- [9] Hagler G S W *et al* 2011 Post-processing method to reduce noise while preserving high time resolution in aethalometer real-time black carbon data *Aerosol Air Qual. Res.* **11** 539–546
- [10] Wong M, Djaja T and Dharmawan B 2018 Cross-calibration approaches for air quality monitoring networks: A review *Sensors* **18** 3785
- [11] Schneider P *et al* 2017 Mapping urban air quality in near real-time using observations from low-cost sensors and model information *Environ. Int.* **106** 234–247
- [12] Korea Ministry of Environment 2018 National ambient air quality standards (Enforcement Decree of the Clean Air Conservation Act) (Sejong, South Korea)
- [13] Chow J C and Watson J G 2002 Review of PM<sub>2.5</sub> and PM<sub>10</sub> apportionment for fossil fuel combustion and other sources by the chemical mass balance receptor model *Energy Fuels* **16** 222–260
- [14] Heal M R, Kumar P and Harrison R M 2012 Particles, air quality, policy and health *Chem. Soc. Rev.* **41** 6606–6630
- [15] Pang Y *et al* 2021 Assessing the representativeness of monitoring stations for air quality in urban areas *Environ. Pollut.* **288** 117753
- [16] Spinelle L, Gerboles M, Villani M G, Aleixandre M and Bonavitacola F 2017 Field calibration of a cluster of low-cost commercially available sensors for air quality monitoring. Part B: NO, CO and CO<sub>2</sub> *Sens. Actuators B Chem.* **238** 706–715
- [17] Page E S 1954 Continuous inspection schemes *Biometrika* **41** 100–115